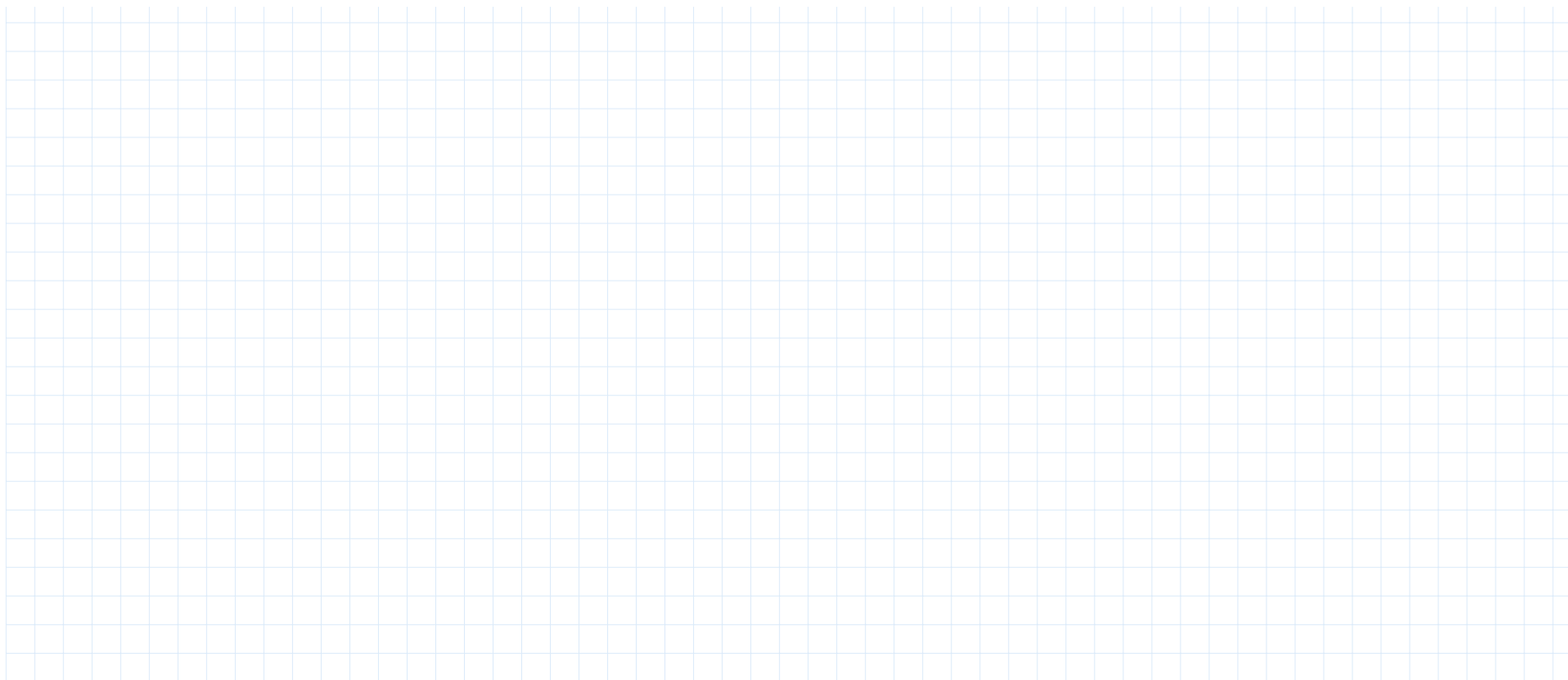


In $y = A \sin(B(x - h)) + D$, the horizontal shift is h and the period is $\frac{2\pi}{|B|}$. Factor the argument before reading the shift.

Example. Rewrite $y = 2 \sin(3x - \pi)$ in shifted form. Find the amplitude, period, phase shift, and range.

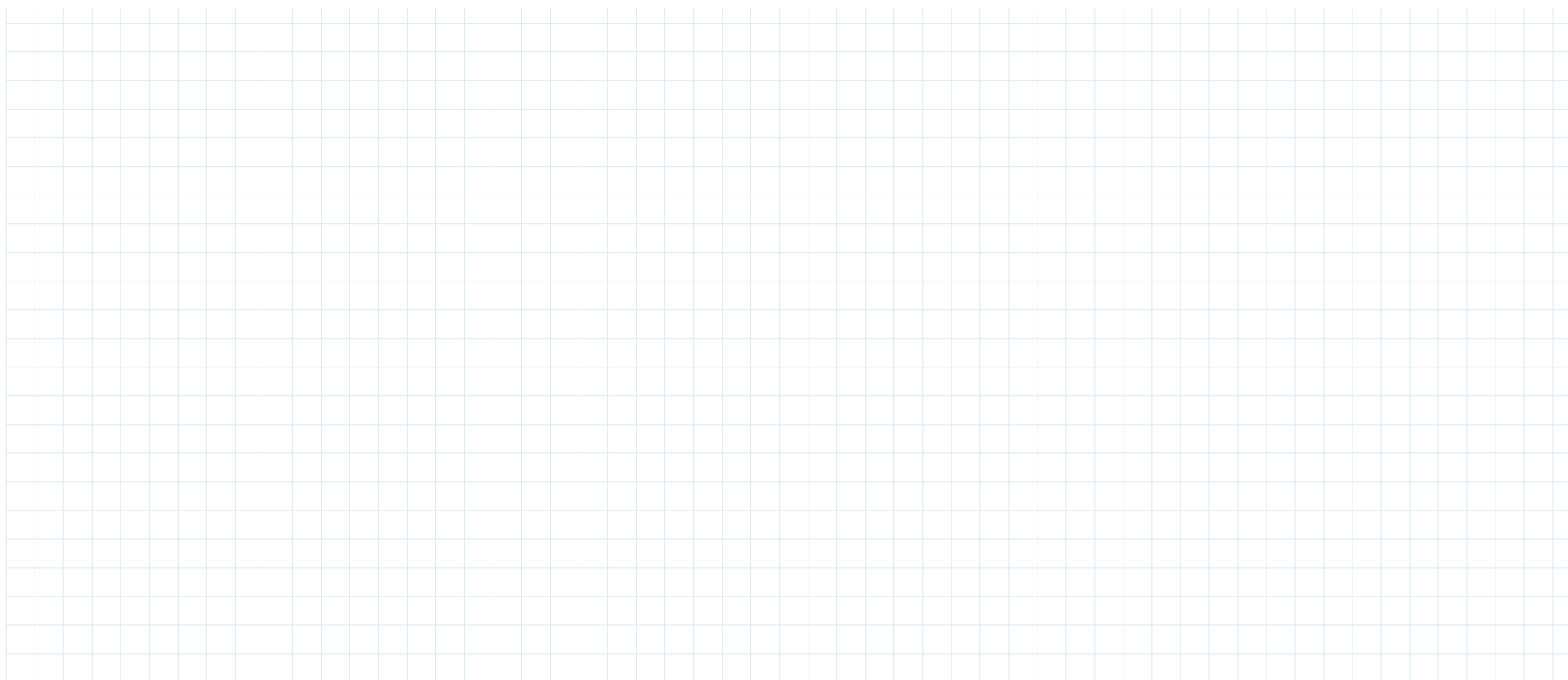


A shifted sine graph

Example. Find the amplitude, period, phase shift, midline, and range of

$$y = 3 \sin(2x - \pi) + 1.$$

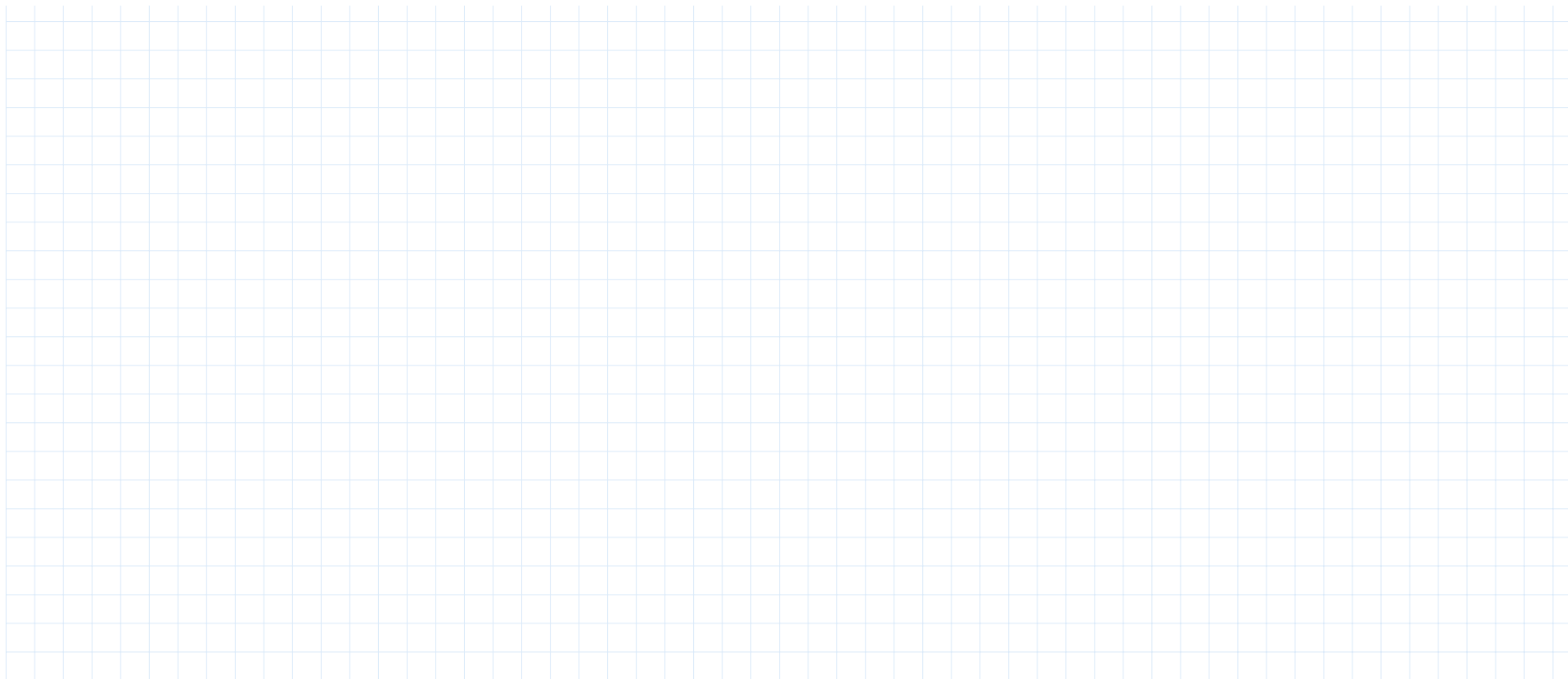
Sketch one cycle and label its five key points.



Example. Sketch one cycle of

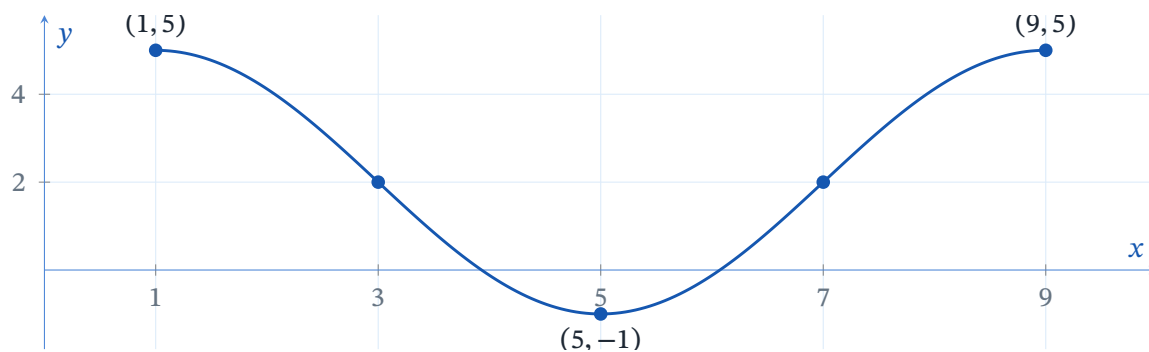
$$y = -2 \cos\left(x + \frac{\pi}{2}\right) + 1.$$

State the amplitude, period, phase shift, midline, range, and five key points.



An equation from a graph

Example. The graph below shows one complete cycle. Write a cosine equation for it. Identify the amplitude, midline, period, and phase shift.

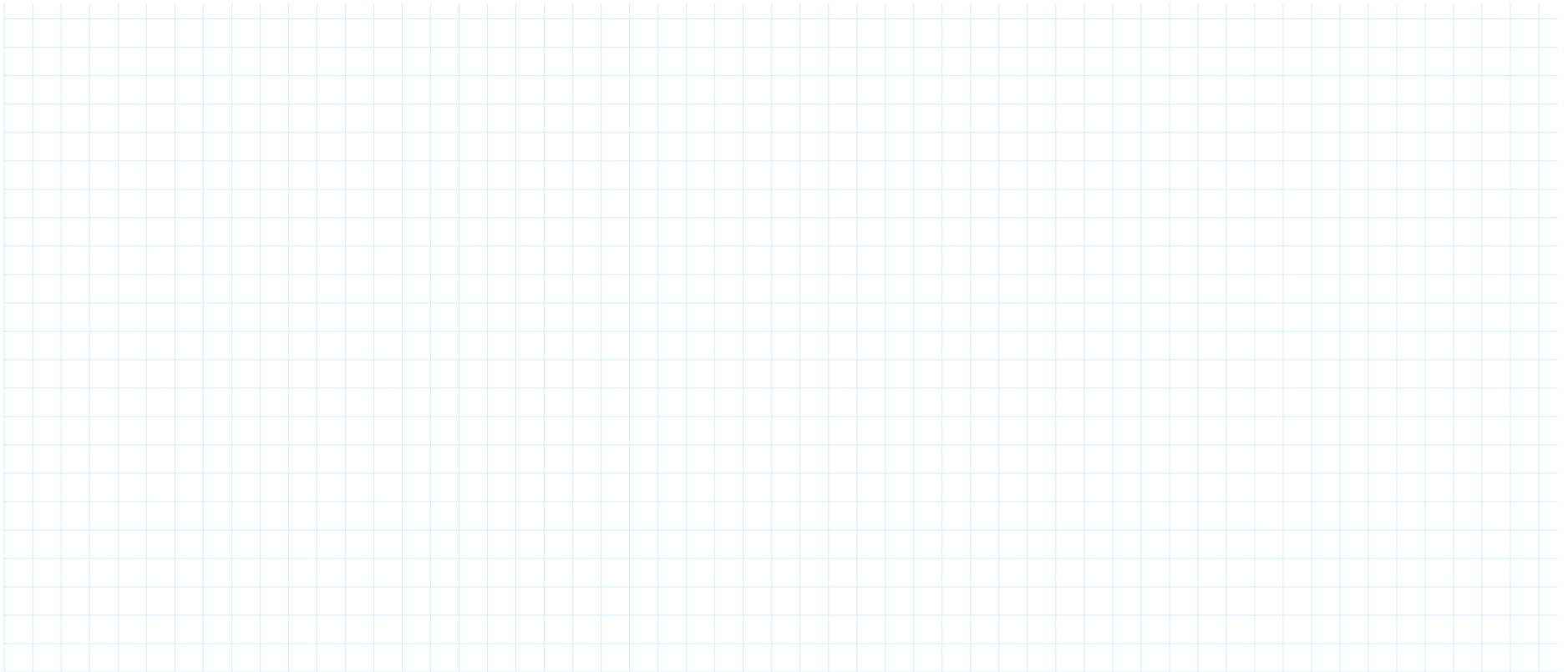


A model from cyclic data

Example. A sensor records the following idealized water depths. Time t is in hours; depth d is in feet.

t	0	3	6	9	12
d	4	10	16	10	4

Assume the data describe one full sinusoidal cycle. Write a model and predict the depth at $t = 4$ hours.



Optional calculator demonstration. A graphing utility fits monthly temperature data with

$$T(m) = 20.8 \sin(0.524m - 2.12) + 54.9,$$

where m is the month number and T is in degrees Fahrenheit. Interpret the amplitude and midline, estimate the period, and predict $T(7)$. Use radian mode.

